

Kaviprakash Ramalingam

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Education

Northeastern University

Master of Science in Information Systems

Sep. 2023 – May 2025

Boston, MA

Experience

Renault Nissan

April 2021 – August 2023

Senior Fullstack Engineer

- Senior developer for Renault's ISIT systems, managing **1,500** global supplier sites across **25** countries.
- Redesigned invoice processing systems for Renault's SDV entity, AMPERE, with **Spring Boot** RESTful APIs for data processing, integrating automated **ETL** pipelines reducing manual touchpoints by **50%** and ensuring consistent data integrity.
- Helped architect a government-access system for Renault Romania by implementing an **OAuth2**-compliant authentication and authorization server, scaling to about **500K** tokens issued daily.
- Engineered performant web applications supporting Renault's EV strategy using **React**, **Tailwind CSS** optimizing **Redux** state management and reducing frontend latency by **40%**.
- Recognized for improving operational efficiency by collaborating with SREs to refine SLIs/SLOs, build cross-platform observability tools and improve service telemetry.

Capgemini Technologies

January 2018 – March 2021

Associate Consultant

- Automated infrastructure provisioning with Terraform and Ansible, cutting setup time by **40%**.
- Implemented **CI/CD** pipelines (**Jenkins**, **AWS**) for microservice deployments, ensuring **100%** automated patching and updates.
- Integrated **Prometheus** and **Grafana** for real-time monitoring, decreasing system downtime and issue resolution time by 35%.
- Implemented a serverless microservices ecosystem using AWS Lambda, API Gateway, and DynamoDB reducing operational costs and improving scalability for a user base of 1 million+.

Projects

Cognitix AI – Multi-Agent Platform | Python, OpenAI, LangChain, Voiceflow, Agentic AI, Prompt Engineering, LLM

February 2025

- Designed a modular LLM tutoring system using **LangChain Agents**, enabling multi-step financial reasoning through controlled tool invocation, memory chaining, and context-aware response planning.
- Integrated real-time market data via **Finnhub API**, and constructed prompt routing logic using persona prompt patterns for generating quizzes, follow-ups, and investment case evaluations based on session-level state.

Rehabilitation Games & Extended Reality (ReGame-XR) | Python, YOLOv8

June 2024

- Developed an AI pipeline in Python to visualize gaze points from raw eye-tracking data, aiding autism behavior analysis
- Trained a YOLOv8 object detection model to analyze gaze patterns, distinguishing autistic and neurotypical individuals.

PULSAR - Digital Transformation | React, Node, MongoDB, SwiftUI, Firebase, Express

September 2023

- Developed microservices-based vehicle monitoring solution using **React** and **Express**, with accessible design (**WCAG**)
- Includes a vehicle health monitoring iOS app leveraging **SwiftUI**, **Firebase**, and **MapKit** for seamless geolocation and dynamic UI
- Implemented layered security with **OAuth 2.0**, **JWT**-based session tokens, and Firebase user verification
- Optimized Redux, implemented server-side rendering and lazy loading on the web; improved iOS interactivity with real-time Firestore and CoreLocation
- Built PWA features, reduced React UI latency by 30% through advanced rendering strategies, virtualization, throttling and memoized components

HORSE & AMPERE – Powertrain | Java, Spring Boot, React, REST, Websockets, Jenkins, Docker, Jest

November 2022

- Built a real-time telemetry dashboard for EV powertrain data using **React** and **WebSockets**, with backend processing via **Spring Boot REST** endpoints and **Redis** caching layers.
- Used **GraphQL** to reduce overfetching for complex vehicle diagnostic reports, improving payload efficiency and UI responsiveness.
- Enabled environment parity using **Docker**-based microservice images across dev, staging, and production, integrated with **Jenkins** CI pipelines, improving rollout reliability and MTTR by 40%.
- Implemented unit and integration tests across tiers, achieving 98% coverage with **Jest**, **React error boundaries**(frontend) and **JUnit** (backend), minimizing fall-outs.

Technical Skills

Languages: Java, JavaScript (ES6+), C++, C#, Python, TypeScript, Swift

Data: SQL, MySQL, OracleDB, MongoDB, PostgreSQL, Redis, Firebase

Frontend: ReactJS, NextJs, Bootstrap, CSS3, HTML5, Socket.IO, Redux, SASS, Tailwind CSS, UIKit, SwiftUI

Cloud& DevOps: AWS EC2, S3, Docker, Kubernetes, Azure, Git, GitHub Actions, Jenkins, Terraform, ELK, Grafana, JIRA, Confluence

Java Ecosystem: Kafka, Hibernate, JPA, MVC, OpenAPI, Spring Boot, Spring Security, Tomcat

Libraries/Frameworks: TensorFlow, PyTorch, NLP, Machine Learning Algorithms, Pandas, NumPy, Scikit-learn, SciPy, Streamlit, Hugging Face Transformers, Selenium, GraphQL, Gradio

LLM Tools: Prompt Engineering, Agentic AI, LangChain, RAG, Transformers, OpenAI, FAISS

iOS Development: Swift, Xcode IDE, Core Data, SwiftUI, iOS Simulator